Cambridge O Level Physics

Paper 1 Multiple Choice · Mark Scheme

MARK SCHEME INFORMATION
Subject: Physics (5054/11) | Session: May/June 2025
Total marks: 40 | One mark per correct answer | No negative marking
This mark scheme is confidential and for examiner use only.

ANSWER KEY

Q No.	Correct Answer
1	D
2	B
3	D
4	B
5	A
6	C
7	A
8	D
9	B
10	B
11	D
12	D
13	C
14	D
15	B
16	D
17	C
18	B
19	C
20	C

Q No.	Correct Answer
21	D
22	B
23	A
24	C
25	D
26	A
27	B
28	D
29	C
30	C
31	D
32	B
33	A
34	A
35	C
36	B
37	C
38	B
39	D
40	C

Cambridge O Level Physics

Paper 1 Multiple Choice · Mark Scheme · Explanatory Notes

EXAMINER NOTES & RATIONALE

Q 1	D	Weight is a force (a vector) as it has both magnitude and direction.
Q 2	B	Distance = area under graph = ½×5×4 + 5×1 + ½×5×4 = 10+5+10 = 25 m.
Q 3	D	The coin is too heavy and falls too short a distance for air resistance to ever equal its
Q 4	B	Point X is the limit of proportionality (beyond Hooke's law still holds to elastic limit).
Q 5	A	Constant density · mass/volume = constant · straight line through origin.
Q 6	C	Centripetal (resultant) force always points toward the centre of the circle.
Q 7	A	Momentum p = mv, unit = kg m/s (equivalent to N s).
Q 8	D	F = impulse/·t = 14000/(70×10$^{\cdot 3}$) = 200 000 N.
Q 9	B	Friction acts left, normal force up, weight down; resultant = net force to right.
Q10	B	Inertia is the resistance to change of motion; determined by mass, not weight.
Q11	D	Pressure = Force/Area; vertical tile has smaller base area · greater pressure · sinks.
Q12	D	Pressure increases with depth; greatest at deepest point in tallest column.
Q13	C	Intermolecular forces in solids are attractive at normal separations.
Q14	D	Higher temperature · faster molecules · more frequent and harder collisions with walls.
Q15	B	Need: heater (energy input), thermometer (·T), balance (mass), stop clock (time).
Q16	D	Fastest molecules escape; average KE of remaining molecules falls · temp drops.
Q17	C	Evaporation rate increases with: higher T, larger A, faster airflow (all three factors).
Q18	B	Free (conduction) electrons, not ions, carry thermal energy rapidly through metals.
Q19	C	Angle of incidence is measured between the incident ray and the normal to the surface.
Q20	C	Mirror need only be half the height of the object for a full-length reflection.
Q21	D	Dull black: good emitter AND good absorber of radiation; poor reflector.
Q22	B	Combination of converging/diverging/converging lenses as shown in ray diagram.
Q23	A	Light in water hitting surface at angle > critical angle · total internal reflection.
Q24	C	Converging lens used as magnifying glass gives a virtual, upright, magnified image.
Q25	D	Sound: longitudinal wave; rarefaction has lower pressure than the compression.
Q26	A	Louder sound = greater amplitude (frequency determines pitch, not loudness).
Q27	B	Two reflected pulses: first returns from crack (closer), second from bottom surface.
Q28	D	I = V/R = 12/100 = 0.12 A; Q = I×t = 0.12×1800 = 216 C.
Q29	C	Conventional current flows + · ·; electrons (negative charge) flow · · +.
Q30	C	Thermistor resistance decreases as temperature increases (NTC characteristic).
Q31	D	Two 6· in series = 12·; 12· parallel 6· = 4·; total = 4+6 = 10·.
Q32	B	Peak voltage and period read directly from oscilloscope divisions shown.
Q33	A	Beta (negative) particles in downward field: force by Fleming's left-hand rule.
Q34	A	Step-up transformer raises voltage for long-distance transmission; requires a.c.